

Group & Ring Theory MATP33: Course Summary of Important Results

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Lecture 0 — Background Results

Theorem 1 (First Isomorphism Theorem). *Let $f : G \rightarrow H$ be a group homomorphism. Then $\ker f \trianglelefteq G$ and $\operatorname{im} f \cong G/\ker f$.*

Theorem 2 (Second Isomorphism Theorem). *Let G be a group, $H \leq G$, and $N \trianglelefteq G$. Then $HN \leq G$, $H \cap N \trianglelefteq H$, and $HN/N \cong H/(H \cap N)$.*

Theorem 3 (Third Isomorphism Theorem). *If $K \trianglelefteq G$ and $K \leq L \trianglelefteq G$, then $(G/K)/(L/K) \cong G/L$.*

Theorem 4 (Correspondence Theorem). *Let $N \trianglelefteq G$. Every subgroup of G/N is of the form H/N where $N \leq H \leq G$, and conversely if $N \leq H \leq G$ then $H/N \leq G/N$.*

Theorem 5 (Lagrange). *If $H \leq G$ and G is finite, then $|H|$ divides $|G|$.*

Theorem 6 (Fundamental Theorem of Finitely Generated Abelian Groups). *Let A be a finitely generated abelian group. Then*

$$A \cong \mathbb{Z}^s \oplus \mathbb{Z}/a_1\mathbb{Z} \oplus \cdots \oplus \mathbb{Z}/a_r\mathbb{Z},$$

where $s \geq 0$ and $a_1 \mid a_2 \mid \cdots \mid a_r$ are non-zero non-units in $\mathbb{Z}$. This decomposition is unique.

Lecture 1

Theorem 7 (1.10). *Let G be a group and X a set. Then:*

- (i) If X is a G -set, then the action of G on X induces a homomorphism $\varphi : G \rightarrow S_X$;*
- (ii) Any homomorphism $\varphi : G \rightarrow S_X$ induces an action of G on X .*

Proof. (i) Define $\varphi : G \rightarrow S_X$ by $(\varphi(a))(x) = a \cdot x$ for $a \in G$ and $x \in X$. We must first check that $\varphi(a) \in S_X$ for each $a \in G$, i.e. that the function $x \mapsto a \cdot x$ is a bijection $X \rightarrow X$.

- *Injective:* Suppose $a \cdot x = a \cdot y$. Then $x = a^{-1} \cdot (a \cdot x) = a^{-1} \cdot (a \cdot y) = y$.
- *Surjective:* For any $x \in X$, we have $a \cdot (a^{-1} \cdot x) = x$, so $x \in \operatorname{im}(\varphi(a))$.

Hence $\varphi(a) \in S_X$. To see φ is a homomorphism, let $a, b \in G$. Then for all $x \in X$,

$$(\varphi(ab))(x) = (ab) \cdot x = a \cdot (b \cdot x) = (\varphi(a))((\varphi(b))(x)) = (\varphi(a)\varphi(b))(x).$$

(ii) Define $a \cdot x := (\varphi(a))(x)$. Then $(ab) \cdot x = (\varphi(ab))(x) = (\varphi(a)\varphi(b))(x) = a \cdot (b \cdot x)$, and $e \cdot x = (\varphi(e))(x) = x$. Hence X is a G -set. $\square$

Theorem 8 (1.11 — Cayley's Theorem). *Every group G is isomorphic to a subgroup of the symmetric group S_G .*

Proof. By Theorem 1.10(i), left translation $a \cdot x := ax$ gives a homomorphism $\varphi : G \rightarrow S_G$. If $a \in \ker \varphi$, then $\varphi(a) = \text{id}_G$, so $ax = x$ for all $x \in G$, giving $a = e$. Hence $\ker \varphi = \{e\}$, and by the First Isomorphism Theorem $G \cong G/\ker \varphi \cong \text{im } \varphi \leq S_G$. $\square$

Definition 1 (Normal core). *Let G be a group and $H \leq G$. The normal core of H in G is*

$$\text{core}(H) := \bigcap_{x \in G} xHx^{-1}.$$

It is the largest normal subgroup of G contained in H .

Theorem 9 (1.12). *Let G be a group and $H \leq G$ a subgroup of index n . Then there is a homomorphism $\varphi : G \rightarrow S_n$ such that $\ker \varphi = \text{core}(H)$.*

Proof. Since $|G/H| = n$, we have $S_{G/H} \cong S_n$. By Theorem 1.10(i), the action of G on the coset space G/H by left multiplication gives a homomorphism $\varphi : G \rightarrow S_n$ with $\varphi(a)(xH) = axH$. Then

$$\begin{aligned} \ker \varphi &= \{a \in G \mid axH = xH \text{ for all } x \in G\} \\ &= \{a \in G \mid x^{-1}ax \in H \text{ for all } x \in G\} \\ &= \{a \in G \mid a \in xHx^{-1} \text{ for all } x \in G\} \\ &= \bigcap_{x \in G} xHx^{-1} = \text{core}(H). \end{aligned} \quad \square$$

Corollary 1 (1.13). *Let G be a group with a normal subgroup H of index n . Then G/H is isomorphic to a subgroup of S_n .*

Proof. Since $H \trianglelefteq G$, conjugation fixes H , so $\text{core}(H) = \bigcap_{x \in G} xHx^{-1} = H$. By Theorem 1.12, $\ker \varphi = H$, and by the First Isomorphism Theorem $G/H \cong \text{im } \varphi \leq S_n$. $\square$

Theorem 10 (1.24 — Orbit-Stabiliser Theorem). *Let G be a group acting on a set X . Then the set of all orbits in X under G is a partition of X . Furthermore, for each $x \in X$ there is a bijection $Gx \rightarrow G/G_x$, and hence*

$$|Gx| = |G : G_x|.$$

Therefore, if X is finite,

$$|X| = \sum_{x \in C} |G : G_x|,$$

where $C \subseteq X$ contains exactly one element from each orbit.

Proof. (Partition.) Define $x \sim y$ if $x = ay$ for some $a \in G$. This is an equivalence relation: reflexivity follows from $e \cdot x = x$, symmetry from $x = ay \Rightarrow y = a^{-1}x$, and transitivity from $x = ay, y = bz \Rightarrow x = (ab)z$. The equivalence class of x is exactly the orbit Gx , so the orbits partition X .

(*Bijection* $Gx \rightarrow G/G_x$.) Define

$$\varphi : Gx \rightarrow G/G_x, \quad ax \mapsto aG_x.$$

The key observation is the following chain of equivalences, for any $a, b \in G$:

$$ax = bx \iff a^{-1}bx = x \iff a^{-1}b \in G_x \iff aG_x = bG_x.$$

- *Well-defined:* If $ax = bx$ (the same orbit element written using two different group elements), then $aG_x = bG_x$, so the coset φ assigns does not depend on the choice of representative.
- *Injective:* If $\varphi(ax) = \varphi(bx)$, i.e. $aG_x = bG_x$, then $ax = bx$ by the chain above.
- *Surjective:* Take any coset $aG_x \in G/G_x$. Since $a \in G$, we have $ax \in Gx$, and $\varphi(ax) = aG_x$.

Hence φ is a bijection, giving $|Gx| = |G/G_x| = |G : G_x|$. Summing over one representative per orbit gives $|X| = \sum_{x \in C} |G : G_x|$. $\square$

Theorem 11 (1.25 — Class Equation). *Let G be a finite group. Then the conjugacy classes partition G , and*

$$|G| = \sum_{a \in C} |G : C_G(a)|,$$

where $C \subseteq G$ contains exactly one element from each conjugacy class.

Proof. Let G act on itself by conjugation, i.e. $a \cdot x = axa^{-1}$. The orbit of x under this action is its conjugacy class $\text{ccl}(x) = \{axa^{-1} \mid a \in G\}$, and the stabiliser of x is $G_x = \{a \in G \mid axa^{-1} = x\} = C_G(x)$. Applying Theorem 1.24 directly gives the partition into orbits and $|G| = \sum_{a \in C} |G : C_G(a)|$. $\square$

Lecture 2

Theorem 12 (2.1). *Let G be a finite group of order p^n , where p is prime and $n \in \mathbb{N}$. Then:*

- (i) G has a non-trivial centre $Z(G)$;
- (ii) $Z(G) \cap K$ is non-trivial for any non-trivial normal subgroup K of G ;
- (iii) if H is a proper subgroup of G , then $H \subsetneq N_G(H)$. In particular, if $|H| = p^{n-1}$ then $H \trianglelefteq G$.

Proof. (i) Since $x \in Z(G)$ iff $\text{ccl}(x) = \{x\}$, the class equation refines to

$$p^n = |Z(G)| + \sum_{x \in C} |G : C_G(x)|,$$

where C runs over conjugacy classes with more than one element. For each $x \in C$, $|C_G(x)|$ properly divides p^n , so $p \mid |G : C_G(x)|$. Hence p divides the sum, and therefore $p \mid |Z(G)|$. Since $e \in Z(G)$, we have $|Z(G)| \geq 1$, so in fact $|Z(G)| \geq p > 1$.

(ii) Intersecting the class decomposition of G with K gives $|K| = |Z(G) \cap K| + \sum_{x \in C} |\text{ccl}(x) \cap K|$. Since $K \trianglelefteq G$, each conjugacy class is either disjoint from K or contained in K , so $p \mid \sum |\text{ccl}(x) \cap K|$. As $p \mid |K|$, we get $p \mid |Z(G) \cap K|$, and the same argument as (i) gives $Z(G) \cap K$ non-trivial.

(iii) Let M be a maximal normal subgroup of G contained in H . Then G/M is a p -group, so by (i) it has a non-trivial centre $L/M \trianglelefteq G/M$. By the Correspondence Theorem, $L \trianglelefteq G$. Since $M \subsetneq L$ strictly, maximality of M forces $L \not\subseteq H$.

It remains to show $L \subseteq N_G(H)$. To show $\ell \in N_G(H)$ we need $\ell^{-1}h\ell \in H$ for all $h \in H$. Take $h \in H$ and $\ell \in L$. Since $\ell M \in L/M = Z(G/M)$, it commutes with every coset, so $(\ell M)(hM) = (hM)(\ell M)$, i.e. $\ell hM = h\ell M$. This gives $\ell^{-1}h\ell \in hM$. Since $h \in H$ and $M \subseteq H$, we have $hM \subseteq H$, so $\ell^{-1}h\ell \in H$. Hence $\ell \in N_G(H)$, so $L \subseteq N_G(H)$.

Since $L \subseteq N_G(H)$ but $L \not\subseteq H$, we have $H \subsetneq N_G(H)$. Finally, if $|H| = p^{n-1}$ then $N_G(H)$ has order strictly greater than p^{n-1} , and the only possibility is $p^n = |G|$, so $N_G(H) = G$ and $H \trianglelefteq G$. $\square$

Theorem 13 (2.2 — Burnside's Lemma). *Let G be a finite group acting on a finite set X , and for $g \in G$ let $X_g := \{x \in X \mid gx = x\}$. Then the number k of orbits in X under G is*

$$k = \frac{1}{|G|} \sum_{g \in G} |X_g|.$$

Proof. Let $S := \{(g, x) \in G \times X \mid gx = x\}$. Fixing g gives $|S| = \sum_{x \in X} |X_g|$; fixing x gives $|S| = \sum_{g \in G} |G_x|$. Let $C \subseteq X$ contain one element from each orbit. By the Orbit-Stabiliser Theorem, $|G_x| = |G|/|Gx| = |G|/|Ga|$ for all $x \in Ga$. Hence

$$\frac{1}{|G|} \sum_{g \in G} |X_g| = \frac{|S|}{|G|} = \frac{1}{|G|} \sum_{x \in X} |G_x| = \frac{1}{|G|} \sum_{a \in C} \sum_{x \in Ga} \frac{|G|}{|Ga|} = \frac{1}{|G|} \sum_{a \in C} |G| = k. \quad \square$$

Remark 1 (Pólya counting). *Burnside's Lemma counts distinct objects under symmetry. As an illustration: how many necklaces with p beads (p prime) can be made from n colours, where two necklaces are the same if one is a rotation of the other?*

Let X be the set of all colourings, so $|X| = n^p$. The cyclic group $G = \langle a \rangle$ of order p acts on X by rotation. For the identity, $X_e = X$ so $|X_e| = n^p$. For any $g \neq e$: since $|G|$ is prime, g generates G , so a necklace fixed by g must be fixed by all rotations, meaning all beads have the same colour. Hence $|X_g| = n$. Burnside then gives

$$k = \frac{1}{p} (n^p + (p-1)n).$$

Lecture 3

Definition 2 (p -group, Sylow p -subgroup). *Let p be a prime. A group G is a p -group if every element of G has order a power of p . If G is finite and $p^m \mid |G|$ but $p^{m+1} \nmid |G|$, then any subgroup of G of order p^m is called a Sylow p -subgroup of G .*

Theorem 14 (3.5 — First Sylow Theorem). *Let G be a finite group and p a prime. If $p^m \mid |G|$, then G has a subgroup of order p^m .*

Proof. If $m = 0$ the result is trivial. We proceed by induction on $n = |G|$, the case $n = 1$ being trivial. Assume the result holds for all groups of order less than n .

Case 1: $p \mid |Z(G)|$. By the Fundamental Theorem of Finitely Generated Abelian Groups, $Z(G)$ contains an element a of order p . Let $C = \langle a \rangle$. Since $a \in Z(G)$, we have $C \trianglelefteq G$, and $|G/C| = n/p$, which is divisible by p^{m-1} . By the induction hypothesis, G/C has a subgroup of order p^{m-1} , which by the Correspondence Theorem is of the form H/C for some $H \leq G$. Then $|H| = |H/C| \cdot |C| = p^{m-1} \cdot p = p^m$.

Case 2: $p \nmid |Z(G)|$. From the class equation $n = |Z(G)| + \sum_a |G : C_G(a)|$, since $p \mid n$ but $p \nmid |Z(G)|$, there must exist some $a \notin Z(G)$ with $p \nmid |G : C_G(a)|$. Since $|G| = |G : C_G(a)| \cdot |C_G(a)|$ and $p^m \mid |G|$ but $p \nmid |G : C_G(a)|$, it follows that $p^m \mid |C_G(a)|$. Since $a \notin Z(G)$, we have $C_G(a) \neq G$, so $|C_G(a)| < n$. By the induction hypothesis $C_G(a)$ contains a subgroup of order p^m , which is also a subgroup of G . $\square$

Corollary 2 (3.6 — Cauchy's Theorem). *If p is a prime dividing $|G|$, then G has an element of order p .*

Proof. By the First Sylow Theorem, G has a subgroup H of order p . Every non-identity element of H has order p . $\square$

Corollary 3 (3.7). *A finite group G is a p -group if and only if $|G|$ is a power of p .*

Proof. If $|G|$ is a power of p , then by Lagrange every element order divides $|G|$, hence is a power of p . Conversely, if $|G|$ is not a power of p , then some prime $q \neq p$ divides $|G|$, and by Cauchy's Theorem G has an element of order q , contradicting that G is a p -group. $\square$

Theorem 15 (3.8 — Second Sylow Theorem). *Let G be a finite group and p a prime. All Sylow p -subgroups of G are conjugate.*

Proof. Let $S, H \in \text{Syl}_p(G)$. Let S act on the coset space G/H by left translation: $s \cdot gH = sgH$.

Orbits have p -power size. By the Orbit-Stabiliser Theorem, each orbit has size $|S : S_{gH}|$, which divides $|S| = p^m$, hence is a power of p .

There must be a fixed point. Since H is a Sylow p -subgroup, $p \nmid |G|/|H| = |G/H|$. Writing $|G/H|$ as a sum of orbit sizes, each of which is a power of p , the only way this sum is not divisible by p is if at least one orbit has size $p^0 = 1$.

A fixed point gives conjugacy. If gH is a fixed point, then $s \cdot gH = gH$ for all $s \in S$, so $g^{-1}sg \in H$ for all $s \in S$. Hence $g^{-1}Sg \subseteq H$, and since $|g^{-1}Sg| = |S| = |H|$, we get $g^{-1}Sg = H$. So S and H are conjugate. $\square$

Corollary 4 (3.10). *A Sylow p -subgroup of G is unique if and only if it is normal in G .*

Proof. If P is the unique Sylow p -subgroup, then $gPg^{-1} = P$ for all $g \in G$ (since conjugates are also Sylow p -subgroups), so $P \trianglelefteq G$. Conversely, if $P \trianglelefteq G$ then P is its own conjugate, so by the Second Sylow Theorem it is the only Sylow p -subgroup. $\square$

Theorem 16 (3.9 — Third Sylow Theorem). *Let G be a finite group and p a prime. The number n_p of Sylow p -subgroups satisfies $n_p \mid |G|$ and $n_p \equiv 1 \pmod{p}$.*

Proof. $n_p \mid |G|$. By the Second Sylow Theorem, all Sylow p -subgroups are conjugate to a fixed $K \in \text{Syl}_p(G)$, so $n_p = |\text{ccl}(K)| = |G : N_G(K)|$, which divides $|G|$.

$n_p \equiv 1 \pmod{p}$. Let $P \in \text{Syl}_p(G)$. Since conjugates of Sylow p -subgroups are again Sylow p -subgroups (by the Second Sylow Theorem), P acts on the set $\text{Syl}_p(G)$ by conjugation:

$$x \cdot Q := xQx^{-1}, \quad x \in P, Q \in \text{Syl}_p(G).$$

This action partitions $\text{Syl}_p(G)$ into orbits. Let $F \subseteq \text{Syl}_p(G)$ denote the set of fixed points (those Q with $xQx^{-1} = Q$ for all $x \in P$), and let $\mathcal{O}$ denote the collection of orbits of size greater than 1. Then:

$$n_p = |\text{Syl}_p(G)| = |F| + \sum_{O \in \mathcal{O}} |O|.$$

By the Orbit-Stabiliser Theorem each $|O|$ divides $|P| = p^m$, so every $O \in \mathcal{O}$ has size p^e for some $e \geq 1$, hence $p \mid |O|$. Therefore $n_p \equiv |F| \pmod{p}$.

It remains to show $F = \{P\}$, which gives $n_p \equiv 1 \pmod{p}$. Clearly $P \in F$. If $Q \in F$, then $xQx^{-1} = Q$ for all $x \in P$, so $P \leq N_G(Q)$. By Corollary 3.10, since $Q \trianglelefteq N_G(Q)$, Q is the unique Sylow p -subgroup of $N_G(Q)$. But P is also a Sylow p -subgroup of $N_G(Q)$, so $P = Q$. Hence $F = \{P\}$. $\square$

Lecture 4 — Modules: the Basics

Definition 3 (4.1 — Left R -module). *Let R be a ring. An abelian group M is a left R -module if there is a map $R \times M \rightarrow M$, $(r, m) \mapsto rm$, such that for all $r, r_1, r_2 \in R$ and $m, m_1, m_2 \in M$:*

$$(i) \quad r(m_1 + m_2) = rm_1 + rm_2;$$

$$(ii) \quad (r_1 + r_2)m = r_1m + r_2m;$$

$$(iii) \quad (r_1r_2)m = r_1(r_2m);$$

$$(iv) \quad 1m = m \text{ if } 1 \in R.$$

Lemma 1 (4.7). *Let M be an R -module. Then for $a \in R$ and $m \in M$:*

$$(i) \quad 0m = 0;$$

$$(ii) \quad a0 = 0;$$

$$(iii) \quad (-a)m = -(am) = a(-m).$$

Proof. (i) $0m = (0 + 0)m = 0m + 0m$. Subtracting $0m$ from both sides gives $0 = 0m$.

(ii) $a0 = a(0 + 0) = a0 + a0$. Subtracting $a0$ from both sides gives $0 = a0$.

(iii) By (i) and axiom (ii): $0 = 0m = (a + (-a))m = am + (-a)m$, so $(-a)m = -(am)$. By (ii) and axiom (i): $0 = a0 = a(m + (-m)) = am + a(-m)$, so $a(-m) = -(am)$. $\square$

Definition 4 (4.8 — Submodule). *A subset N of an R -module M is an R -submodule of M if:*

$$(i) \quad N \neq \emptyset;$$

$$(ii) \quad a - b \in N \text{ for all } a, b \in N;$$

$$(iii) \quad ra \in N \text{ for all } a \in N, r \in R.$$

Theorem 17 (4.12). *Let $(N_i)_{i \in I}$ be a family of R -submodules of an R -module M . Then $\bigcap_{i \in I} N_i$ is also an R -submodule of M .*

Proof. Since $0 \in N_i$ for all i , we have $0 \in \bigcap N_i$, so it is non-empty. Let $x, y \in \bigcap N_i$ and $r \in R$. For each i , since N_i is a submodule, $x - y \in N_i$ and $rx \in N_i$. Hence $x - y$ and rx lie in $\bigcap N_i$. $\square$

Definition 5 (4.13 — Generated submodule, cyclic module). *Let $S \subseteq M$. The family $\mathcal{A} = \{N \mid N \text{ is an } R\text{-submodule of } M \text{ containing } S\}$ is non-empty (since $M \in \mathcal{A}$), so by Theorem 4.12 the intersection $\bigcap_{N \in \mathcal{A}} N$ is an R -submodule. It is the smallest submodule of M containing S , called the submodule generated by S and denoted (S) . If $S = \{x_1, \dots, x_n\}$ we write $(x_1, \dots, x_n)$. An R -module M is cyclic if $M = (x)$ for some $x \in M$.*

Theorem 18 (4.15). *Let R be a ring with unity and $x_1, \dots, x_n \in M$. Then*

$$(x_1, \dots, x_n) = \sum_{i=1}^n Rx_i := \{r_1x_1 + \dots + r_nx_n \mid r_i \in R\}.$$

Proof. $(\supseteq)$ The submodule $(x_1, \dots, x_n)$ contains each x_i . By closure under scalar multiplication it contains rx_i for all $r \in R$, and by closure under addition it contains every sum $r_1x_1 + \dots + r_nx_n$. Hence $\sum Rx_i \subseteq (x_1, \dots, x_n)$.

$(\subseteq)$ We show $\sum Rx_i$ is a submodule containing each x_i ; then $(x_1, \dots, x_n) \subseteq \sum Rx_i$ follows from minimality of $(x_1, \dots, x_n)$.

Each $x_i \in \sum Rx_i$ since $x_i = 1 \cdot x_i$ (using unity). For the submodule conditions: $0 = \sum 0 \cdot x_i \in \sum Rx_i$, so it is non-empty. For $\sum r_ix_i, \sum s_ix_i \in \sum Rx_i$ and $r \in R$,

$$\sum r_ix_i - \sum s_ix_i = \sum (r_i - s_i)x_i \in \sum Rx_i, \quad r \sum r_ix_i = \sum (rr_i)x_i \in \sum Rx_i. \quad \square$$

Definition 6 (4.18 — Sum of submodules). *Let $N_1, \dots, N_k$ be R -submodules of M . Their sum $\sum_{i=1}^k N_i$ is the submodule of M generated by $\bigcup_{i=1}^k N_i$, i.e. the smallest submodule of M containing each N_i .*

Lecture 5 — Modules: continuation

Theorem 19 (4.19). *Let $N_1, \dots, N_k$ be R -submodules of an R -module M . Then*

$$\sum_{i=1}^k N_i = \{x_1 + \dots + x_k \mid x_i \in N_i\}.$$

Proof. Let

$$A = \{x_1 + \dots + x_k \mid x_i \in N_i\}.$$

It is straightforward to check that A is a submodule of M , and it contains every N_i . Hence, by minimality,

$$\sum_{i=1}^k N_i \subseteq A.$$

Conversely, $\sum_{i=1}^k N_i$ contains every N_i and is closed under addition, so it contains every element of A . Therefore

$$A \subseteq \sum_{i=1}^k N_i.$$

□

Definition 7 (4.21 — Direct sum). *The sum $N_1 + \cdots + N_k$ is direct if every element has a unique representation*

$$x = x_1 + \cdots + x_k, \quad x_i \in N_i.$$

In this case we write

$$N_1 \oplus \cdots \oplus N_k.$$

Theorem 20 (4.22). *Let $N_1, \dots, N_k$ be submodules of an R -module M . The following are equivalent:*

- (i) $\sum_{i=1}^k N_i$ is direct;
- (ii) if $\sum_{i=1}^k x_i = 0$, where $x_i \in N_i$, then every $x_i = 0$;
- (iii) for every i ,

$$N_i \cap \sum_{j \neq i} N_j = \{0\}.$$

Proof. (i) $\Rightarrow$ (ii): Zero already has the representation $0 + \cdots + 0$, so uniqueness implies that every $x_i = 0$.

(ii) $\Rightarrow$ (iii): If

$$x \in N_i \cap \sum_{j \neq i} N_j,$$

then $x = \sum_{j \neq i} x_j$ for some $x_j \in N_j$. Hence

$$x - \sum_{j \neq i} x_j = 0,$$

so (ii) gives $x = 0$.

(iii) $\Rightarrow$ (i): Suppose

$$\sum_{i=1}^k x_i = \sum_{i=1}^k y_i, \quad x_i, y_i \in N_i.$$

For every i ,

$$x_i - y_i = - \sum_{j \neq i} (x_j - y_j) \in N_i \cap \sum_{j \neq i} N_j = \{0\}.$$

Thus $x_i = y_i$ for every i , so the representation is unique.

□

Definition 8 (4.23 — *R*-homomorphism). Let M and N be *R*-modules. A map $f: M \rightarrow N$ is an *R*-homomorphism if, for all $x, y \in M$ and $r \in R$,

$$f(x + y) = f(x) + f(y), \quad f(rx) = rf(x).$$

The set of all *R*-homomorphisms $M \rightarrow N$ is denoted

$$\text{Hom}_R(M, N).$$

An *R*-homomorphism $M \rightarrow M$ is called an endomorphism, and

$$\text{End}_R(M) := \text{Hom}_R(M, M).$$

Theorem 21 (4.30). Let $M = \bigoplus_{i=1}^k M_i$ be a direct sum of *R*-modules. Then

$$\text{Hom}_R(M, M) \cong (\text{Hom}_R(M_j, M_i))_{1 \leq i, j \leq k}$$

as rings, where the right-hand side is a ring T of $k \times k$ matrices under the usual matrix addition and multiplication, with (i, j) entry in $\text{Hom}_R(M_j, M_i)$.

Proof. Let $\lambda_j: M_j \rightarrow M$ be the natural inclusion and $\pi_i: M \rightarrow M_i$ the natural projection. Two key identities hold:

$$\sum_{\ell=1}^k \lambda_\ell \pi_\ell = \text{id}_M, \quad \pi_p \lambda_q = \delta_{pq} \text{id}_{M_p}.$$

The first says every $x \in M$ equals the sum of its components; the second says projecting the q -th inclusion gives zero unless $p = q$.

Define $\sigma: \text{Hom}_R(M, M) \rightarrow T$ by $\sigma(\phi) = (\pi_i \phi \lambda_j)_{ij}$.

Ring homomorphism. Additivity is immediate. For multiplication, using the first identity:

$$(\sigma(\phi)\sigma(\psi))_{ij} = \sum_{\ell} \pi_i \phi \lambda_\ell \pi_\ell \psi \lambda_j = \pi_i \phi \left(\sum_{\ell} \lambda_\ell \pi_\ell \right) \psi \lambda_j = \pi_i \phi \text{id}_M \psi \lambda_j = (\sigma(\phi\psi))_{ij}.$$

Injective. Suppose $\sigma(\phi) = 0$, so $\pi_i \phi \lambda_j = 0$ for all i, j . Applying λ_i on the left and summing over i using the first identity:

$$\phi \lambda_j = \text{id}_M \cdot \phi \lambda_j = \left(\sum_i \lambda_i \pi_i \right) \phi \lambda_j = \sum_i \lambda_i (\pi_i \phi \lambda_j) = 0.$$

Now applying π_j on the right and summing over j :

$$\phi = \phi \cdot \text{id}_M = \phi \left(\sum_j \lambda_j \pi_j \right) = \sum_j (\phi \lambda_j) \pi_j = 0.$$

Surjective. Given $f = (f_{ij}) \in T$, set $\phi = \sum_{i,j} \lambda_i f_{ij} \pi_j$. Using the second identity:

$$(\sigma(\phi))_{st} = \pi_s \phi \lambda_t = \sum_{i,j} \pi_s \lambda_i f_{ij} \pi_j \lambda_t = \sum_{i,j} \delta_{si} f_{ij} \delta_{jt} = f_{st}.$$

Hence $\sigma(\phi) = f$, so σ is surjective. □

Definition 9 (Quotient module). Let N be an R -submodule of an R -module M . The set of cosets $M/N = \{a + N \mid a \in M\}$ becomes an R -module under

$$(a + N) + (b + N) = (a + b) + N, \quad r(a + N) = ra + N.$$

It is called the quotient module of M by N .

Theorem 22 (4.31 — Correspondence Theorem for Modules). Let N be an R -submodule of an R -module M . Every R -submodule of M/N is of the form U/N , where U is an R -submodule of M containing N .

Proof. The canonical projection $f : M \rightarrow M/N$, $f(x) = x + N$, is an R -homomorphism:

$$f(x+y) = (x+y)+N = (x+N)+(y+N) = f(x)+f(y), \quad f(rx) = rx+N = r(x+N) = rf(x).$$

Let X be an R -submodule of M/N . The structure of the proof is: define U as the preimage of X under f , show U is a submodule of M containing N , and conclude $X = U/N$.

Set $U = f^{-1}(X) = \{x \in M \mid f(x) \in X\}$.

$N \subseteq U$. For any $x \in N$, $f(x) = x + N = 0 + N = 0_{M/N} \in X$ (since X is a submodule, it contains 0). Hence $x \in U$.

U is an R -submodule. Since $0 \in N \subseteq U$, U is non-empty. For $x, y \in U$ and $r \in R$, since f is an R -homomorphism:

$$f(x - y) = f(x) - f(y) \in X, \quad f(rx) = rf(x) \in X,$$

so $x - y \in U$ and $rx \in U$.

$X = U/N$. By definition of U , $f(U) \subseteq X$. For the reverse inclusion, take any $x \in X$; since f is surjective there exists $y \in M$ with $f(y) = x$, so $y \in U$ and $x \in f(U)$. Hence $X = f(U) = U/N$. $\square$

Theorem 23 (4.32 — First Isomorphism Theorem for Modules). Let $f : M \rightarrow N$ be an R -homomorphism. Then $M/\ker f \cong f(M)$.

Proof. Define $g : M/\ker f \rightarrow f(M)$ by $g(x + \ker f) = f(x)$. We know from group theory that g is an isomorphism of abelian groups (well-defined and injective because $x + \ker f = y + \ker f \iff x - y \in \ker f \iff f(x) = f(y)$; surjective onto $f(M)$ by definition). It remains to check that g respects the R -action:

$$g(r(x + \ker f)) = g(rx + \ker f) = f(rx) = rf(x) = rg(x + \ker f). \quad \square$$

Theorem 24 (Second Isomorphism Theorem for Modules). Let U and V be R -submodules of an R -module M . Then $(U + V)/V \cong U/(U \cap V)$.

Proof. Define $f : U \rightarrow (U + V)/V$ by $f(u) = u + V$.

f is an R -homomorphism. For $u_1, u_2 \in U$ and $r \in R$:

$$f(u_1 + u_2) = (u_1 + u_2) + V = (u_1 + V) + (u_2 + V) = f(u_1) + f(u_2),$$

$$f(ru) = ru + V = r(u + V) = rf(u).$$

f is surjective. Any element of $(U + V)/V$ has the form $(u + v) + V = u + V = f(u)$ for some $u \in U$, $v \in V$ (since $v \in V$ means $v + V = V$).

$\ker f = U \cap V$. We have $f(u) = 0$ iff $u + V = V$ iff $u \in V$, so $\ker f = \{u \in U \mid u \in V\} = U \cap V$.

By the First Isomorphism Theorem, $U/(U \cap V) \cong (U + V)/V$. $\square$

Theorem 25 (Third Isomorphism Theorem for Modules). *Let $U \subseteq V$ be R -submodules of an R -module M . Then $(M/U)/(V/U) \cong M/V$.*

Proof. Define $f : M/U \rightarrow M/V$ by $f(x+U) = x+V$.

f is well-defined. If $x+U = y+U$ then $x-y \in U \subseteq V$, so $x+V = y+V$.

f is an R -homomorphism. For $x+U, y+U \in M/U$ and $r \in R$:

$$f((x+U)+(y+U)) = f((x+y)+U) = (x+y)+V = (x+V)+(y+V) = f(x+U)+f(y+U),$$

$$f(r(x+U)) = f(rx+U) = rx+V = r(x+V) = rf(x+U).$$

f is surjective. For any $x+V \in M/V$, $f(x+U) = x+V$.

$\ker f = V/U$. We have $f(x+U) = 0$ iff $x+V = V$ iff $x \in V$, so $\ker f = \{x+U \mid x \in V\} = V/U$.

By the First Isomorphism Theorem, $(M/U)/(V/U) \cong M/V$. $\square$

Theorem 26 (4.33). *Let $A \leq M$ and $B \leq N$ be R -submodules. Then $(M \times N)/(A \times B) \cong M/A \times N/B$.*

Proof. Define $f : M \times N \rightarrow M/A \times N/B$ by $f(m, n) = (m+A, n+B)$. This is a surjective R -homomorphism, and $\ker f = \{(m, n) \mid m \in A, n \in B\} = A \times B$. By the First Isomorphism Theorem, $(M \times N)/(A \times B) \cong M/A \times N/B$. $\square$

Lecture 6 — Simple, Completely Reducible, and Free Modules

Definition 10 (5.1 — Simple module). *Let R be a ring with unity. An R -module M is called simple (or irreducible) if $M \neq 0$ and the only R -submodules of M are 0 and M .*

Theorem 27 (5.5 — Characterisation of Simple Modules). *Let R be a ring with unity and M an R -module. The following are equivalent:*

- (i) M is simple;
- (ii) $M \neq 0$, and M is generated by any $0 \neq x \in M$;
- (iii) $M \cong R/I$, where I is a maximal left ideal of R .

Proof. (i) $\Rightarrow$ (ii): Let $0 \neq x \in M$. Then Rx is a non-zero submodule of M , so $Rx = M$ by simplicity.

(ii) $\Rightarrow$ (iii): Take $0 \neq x \in M$ and define $f : R \rightarrow M$ by $f(r) = rx$. This is a surjective R -homomorphism, so letting $I = \ker f$, the First Isomorphism Theorem gives $R/I \cong M$. Note that (ii) implies M is simple: if $0 \neq N \leq M$ and $0 \neq x \in N$, then $M = Rx \subseteq N$. Hence $R/I \cong M$ is simple. By the Correspondence Theorem (4.31), any left ideal J with $I \subseteq J \subseteq R$ corresponds to a submodule J/I of R/I . Simplicity forces $J/I = 0$ or $J/I = R/I$, i.e. $J = I$ or $J = R$, so I is maximal.

(iii) $\Rightarrow$ (i): Let $0 \neq N \leq M \cong R/I$. By the Correspondence Theorem (4.31), $N \cong J/I$ for some left ideal J with $I \subsetneq J \subseteq R$. Since I is maximal, $J = R$, so $N = M$. $\square$

Theorem 28 (5.6 — Schur's Lemma). *Let M be a simple R -module. Then $\text{Hom}_R(M, M)$ is a division ring.*

Proof. Let $0 \neq f \in \text{Hom}_R(M, M)$. We show f is invertible and $f^{-1} \in \text{Hom}_R(M, M)$.

f is bijective. Since $\ker f$ and $\text{im } f$ are submodules of M , simplicity forces each to be 0 or M . As $f \neq 0$, we have $\ker f \neq M$ and $\text{im } f \neq 0$, so $\ker f = 0$ (injective) and $\text{im } f = M$ (surjective).

f^{-1} is an R -homomorphism. We cannot push f^{-1} through operations directly, but we can push f through them. The strategy is: to prove $f^{-1}(u) = v$, it suffices to show $f(u) = f(v)$, since f is injective. For $x, y \in M$ and $r \in R$:

$$\begin{aligned} f(f^{-1}(x + y)) &= x + y = f(f^{-1}(x)) + f(f^{-1}(y)) = f(f^{-1}(x) + f^{-1}(y)), \\ f(f^{-1}(rx)) &= rx = rf(f^{-1}(x)) = f(rf^{-1}(x)), \end{aligned}$$

where we used that f is itself an R -homomorphism in the last step of each line. Injectivity of f gives $f^{-1}(x + y) = f^{-1}(x) + f^{-1}(y)$ and $f^{-1}(rx) = rf^{-1}(x)$. $\square$

Definition 11 (5.7 — Completely reducible module). Let R be a ring with unity. An R -module M is called completely reducible (or semisimple) if $M = \sum_{\alpha \in \Lambda} M_\alpha$, where each M_α is a simple R -submodule.

Theorem 29 (5.9). Let $M = \sum_{\alpha \in \Lambda} M_\alpha$ be a sum of simple R -submodules, and let K be a submodule of M . Then there exists $\Lambda' \subseteq \Lambda$ such that $M = K \oplus (\bigoplus_{\alpha \in \Lambda'} M_\alpha)$.

Proof. Let S be the family of subsets $A \subseteq \Lambda$ satisfying two conditions: $\bigoplus_{\alpha \in A} M_\alpha$ is a direct sum, and $K \cap \bigoplus_{\alpha \in A} M_\alpha = 0$. Together these mean $K \oplus \bigoplus_{\alpha \in A} M_\alpha$ is a genuine direct sum. Since $\emptyset$ satisfies both conditions trivially, $S \neq \emptyset$. Partially order S by inclusion; every chain (A_i) in S has upper bound $\bigcup A_i$ in S . By Zorn's Lemma, S has a maximal element A .

Set $N = K \oplus \bigoplus_{\alpha \in A} M_\alpha$. We claim $N = M$. Fix any $\beta \in \Lambda$. Since M_β is simple, $M_\beta \cap N$ is either 0 or M_β . If $M_\beta \cap N = 0$, then M_β intersects neither K nor $\bigoplus_{\alpha \in A} M_\alpha$, so $A \cup \{\beta\}$ would satisfy both conditions in S , contradicting maximality of A . Hence $M_\beta \cap N = M_\beta$, i.e. $M_\beta \subseteq N$, for every $\beta \in \Lambda$. Since $M = \sum_{\beta} M_\beta$, we conclude $N = M$. $\square$

Corollary 5 (5.10). If $M = \sum_{\alpha \in \Lambda} M_\alpha$ is a sum of simple R -submodules, then there exists $\Lambda' \subseteq \Lambda$ such that $M = \bigoplus_{\alpha \in \Lambda'} M_\alpha$.

Proof. Apply Theorem 5.9 with $K = \{0\}$. $\square$

Definition 12 (5.11 — Linear independence and basis). A sequence $x_1, \dots, x_n$ in an R -module M is linearly independent if $\sum_{i=1}^n a_i x_i = 0$ implies $a_1 = \dots = a_n = 0$ for any $a_i \in R$. A subset $S \subseteq M$ is linearly independent if every finite sequence of distinct elements of S is linearly independent.

A subset $B \subseteq M$ is a basis of M if B generates M and B is linearly independent. An R -module that has a basis is called a free module.

Lecture 7 — Free Modules and Chain Conditions

Theorem 30 (5.19). Let M be a free R -module with basis $\{f_1, \dots, f_n\}$. Then $M \cong R^n$ as R -modules.

Proof. Define $\varphi : M \rightarrow R^n$ by $\varphi(\sum_{i=1}^n r_i f_i) = (r_1, \dots, r_n)$, i.e. read off the coordinates in the basis. Since every element of M has a unique representation as $\sum r_i f_i$ (by linear independence of the f_i), φ is well-defined. It is clearly an R -homomorphism, injective (if $(r_1, \dots, r_n) = 0$ then each $r_i = 0$, so $m = 0$), and surjective (every $(r_1, \dots, r_n) \in R^n$ is the image of $\sum r_i f_i$). $\square$

Theorem 31 (5.20). *Let M be a finitely generated free R -module. Then every basis of M is finite.*

Proof. Let $\{e_i\}_{i \in \Lambda}$ be a basis and $\{x_1, \dots, x_n\}$ a finite generating set. Since elements of a module are finite sums by definition, each x_j involves only finitely many nonzero coefficients, so is supported on a finite subset $S_j \subseteq \Lambda$. Every basis element must appear in the expression of some generator (otherwise it could not be reached), so $\Lambda \subseteq S_1 \cup \dots \cup S_n$, a finite union of finite sets. Hence Λ is finite. $\square$

Theorem 32 (5.21). *Let M be a finitely generated free module over a commutative ring R . Then all bases of M have the same number of elements.*

Proof. Suppose M has two bases of sizes m and n respectively. By Theorem 5.19, $M \cong R^m$ and $M \cong R^n$, so $R^m \cong R^n$. It suffices to show $m = n$.

Assume for contradiction $m < n$. Let $\phi : R^m \rightarrow R^n$ be an R -isomorphism with inverse $\psi = \phi^{-1}$. Let $(e_1, \dots, e_m)$ and $(f_1, \dots, f_n)$ be the standard bases of R^m and R^n . Since ϕ is an R -homomorphism, it is determined by where it sends each e_i , and each image lands in R^n , so we can write

$$\phi(e_i) = \sum_{j=1}^n a_{ji} f_j, \quad \psi(f_j) = \sum_{k=1}^m b_{kj} e_k.$$

Let $A = (a_{ji})$ be the resulting $n \times m$ matrix and $B = (b_{kj})$ the $m \times n$ matrix. Since $\psi\phi = \text{id}_{R^m}$ and $\phi\psi = \text{id}_{R^n}$, substituting the expressions above and using linear independence of the bases gives $BA = I_m$ and $AB = I_n$.

Pad these to $n \times n$ matrices by setting $A' = \begin{pmatrix} A & 0 \end{pmatrix}$ and $B' = \begin{pmatrix} B \\ 0 \end{pmatrix}$. Then $A'B' = I_n$, so $\det(A'B') = 1$. But $B'A' = \begin{pmatrix} I_m & 0 \\ 0 & 0 \end{pmatrix}$, so $\det(B'A') = 0$. Since R is commutative, determinants multiply, giving

$$1 = \det(A'B') = \det(A') \det(B') = \det(B'A') = 0,$$

a contradiction. By symmetry $n < m$ also fails, so $m = n$. $\square$

Definition 13 (5.22 — Rank). *The number of elements in any basis of a finitely generated free module M over a commutative ring R is called the rank of M , written $\text{rk}(M)$. If R is a field, the rank is called the dimension.*

Lemma 2 (5.24). *Every finitely generated R -module is a homomorphic image of a finitely generated free module.*

Proof. Let M be generated by $x_1, \dots, x_n$. Define $\phi : R^n \rightarrow M$ by $\phi(r_1, \dots, r_n) = \sum_{i=1}^n r_i x_i$. This is a surjective R -homomorphism (surjective since the x_i generate M). The generators x_i may satisfy relations, so ϕ need not be injective — but by the First Isomorphism Theorem $M \cong R^n / \ker \phi$, expressing M as a quotient of the free module R^n . $\square$

Lemma 3 (5.25). *Let R be a PID and F a free R -module of rank n . Then every submodule $K \leq F$ is free of rank $m \leq n$.*

Proof. By Theorem 5.19, $F \cong R^n$, so we work with $K \leq R^n$ and induct on n . The base case $n = 0$ is trivial since $K = 0$ is free on the empty set.

Define $\pi : R^n \rightarrow R$ by $\pi(x_1, \dots, x_n) = x_1$, and restrict it to K .

Case 1: $\pi|_K = 0$. Every element of K has first coordinate zero, so $K \leq \{0\} \times R^{n-1} \cong R^{n-1}$. By induction K is free of rank $\leq n - 1 \leq n$.

Case 2: $\pi|_K \neq 0$. The image $\pi(K)$ is a nonzero ideal of R . Since R is a PID, $\pi(K) = Ra$ for some $a \neq 0$. Pick $k \in K$ with $\pi(k) = a$.

We claim $K = Rk \oplus \ker(\pi|_K)$, where $\ker(\pi|_K) = \{x \in K \mid x_1 = 0\}$ is the set of elements of K with first coordinate zero, so $\ker(\pi|_K) \leq \{0\} \times R^{n-1} \cong R^{n-1}$.

$K = Rk + \ker(\pi|_K)$: For any $x \in K$, write $\pi(x) = ba$ for some $b \in R$. Then $\pi(x - bk) = ba - ba = 0$, so $x - bk \in \ker(\pi|_K)$, and $x = bk + (x - bk)$.

The sum is direct: If $ck \in \ker(\pi|_K)$, then $0 = \pi(ck) = ca$. Since R is a PID (hence an integral domain) and $a \neq 0$, we get $c = 0$.

By induction $\ker(\pi|_K)$ is free of rank $s \leq n - 1$, and $Rk \cong R$ is free of rank 1. Hence $K \cong R^s \oplus R = R^{s+1}$, which has rank $m = s + 1 \leq n$. $\square$

Lecture 8 — Noetherian and Artinian Modules

Definition 14 (6.1 — Noetherian and Artinian modules). *An R -module M is called Noetherian if every ascending chain of submodules*

$$M_1 \subseteq M_2 \subseteq M_3 \subseteq \dots$$

stabilises, i.e. there exists k such that $M_k = M_{k+1} = \dots$. This is called the ascending chain condition (ACC). An R -module M is called Artinian if every descending chain of submodules stabilises (descending chain condition, DCC).